

09

Projection

Putting Orthogonal Projection and Orthogonal Matrices into Practice

This chapter is a continuation of the previous chapter's Section 5 — Orthogonal Projection. Orthogonal matrices play an important role in orthogonal projection, and an orthogonal matrix is itself an orthonormal basis. We will also encounter orthogonal matrices in topics such as regression, spectral decomposition, singular value decomposition, and principal component analysis. This chapter begins by understanding 2×2 and 3×3 orthogonal matrices from a geometric point of view, then introduces various properties of orthogonal matrices, and finally covers the application of orthogonal projection in Gram-Schmidt orthogonalization and QR decomposition.

9.1 2×2 Orthogonal Matrices



Core skills covered in this section:

- ▶ The identity matrix, rotation matrices, reflection matrices, permutation matrices, and combinations of them are all orthogonal matrices.
- ▶ The identity matrix represents the identity transformation with respect to the standard orthonormal basis.
- ▶ Using orthogonal projection to obtain vector projections.
- ▶ In the plane, the sum of the orthogonally projected vectors can losslessly recover the original vector.
- ▶ The orthogonal complement contains all the vectors perpendicular to the original space.

The identity matrix, rotation matrices, reflection matrices, permutation matrices, and their composites are all orthogonal matrices. Orthogonal matrices are widely used in orthogonal projection, and we will see orthogonal matrices applied throughout various common geometric decompositions. This section starts with two examples — the identity matrix and the rotation matrix — to understand 2×2 orthogonal matrices from a planar geometric point of view.

The 2×2 Identity Matrix

Let's first look at an extremely simple matrix multiplication.

The identity matrix I times the column vector x gives back x , i.e.,

$$\mathbf{I}\mathbf{x} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \mathbf{x}$$

Behind this "simple" matrix multiplication lies something far more interesting!

Viewed from the third perspective of matrix multiplication — linear combination of column vectors — can be written as

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = x_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$\mathbf{e}_1 \qquad \mathbf{e}_2$

As Figure 1 shows, (x_1, x_2) are the coordinates of the vector $\mathbf{x}$ in the standard orthonormal basis $[\mathbf{e}_1, \mathbf{e}_2]$.

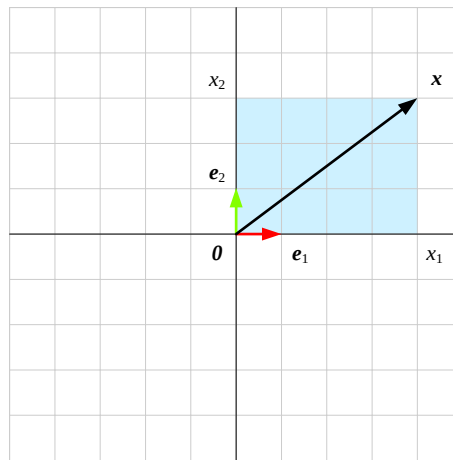


Figure 1. The Plane Spanned by Linear Combinations of $[\mathbf{e}_1, \mathbf{e}_2]$

From another angle, provided the shapes are compatible, matrix multiplication satisfies the distributive property, i.e.,

$$\begin{aligned} (\mathbf{A} + \mathbf{B})\mathbf{C} &= \mathbf{AC} + \mathbf{BC} \\ \mathbf{A}(\mathbf{B} + \mathbf{C}) &= \mathbf{AB} + \mathbf{AC} \end{aligned}$$

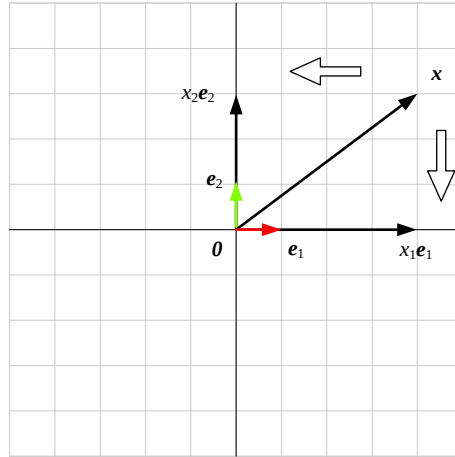
Based on the first equality in , can be written as

$$\left(\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right) \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

Expanding using the distributive property of matrix multiplication

$$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

As Figure 2 shows, we find that this is the orthogonal projection of $\mathbf{x}$ in the standard orthonormal basis $[\mathbf{e}_1, \mathbf{e}_2]$.

Figure 2. Orthogonal Projection of the Vector $\mathbf{x}$ in the Standard Orthonormal Basis $[\mathbf{e}_1, \mathbf{e}_2]$

Orthogonal Projection Toward the $\mathbf{e}_1$ Direction

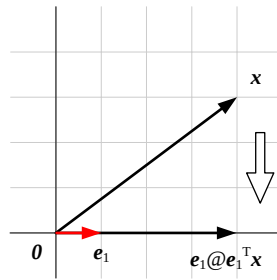
How does this relate to the orthogonal projection introduced in Section 5 of the previous chapter?

Projecting the column vector $\mathbf{x}$ onto the direction vector (unit vector) $\mathbf{e}_1$, the corresponding projection matrix $\mathbf{P}_1$ is

$$\mathbf{P}_1 = \mathbf{e}_1 @ \mathbf{e}_1^T = \begin{bmatrix} 1 \\ 0 \end{bmatrix} @ \begin{bmatrix} 1 \\ 0 \end{bmatrix}^T = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$

As Figure 3 shows, the projection result corresponds to the following matrix multiplication

$$\mathbf{P}_1 \mathbf{x} = \underbrace{\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}}_{\mathbf{e}_1 @ \mathbf{e}_1^T} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_1 \\ 0 \end{bmatrix}$$

Figure 3. Orthogonal Projection of the Two-Dimensional Column Vector $\mathbf{x}$ Onto the $\mathbf{e}_1$ Direction

Orthogonal Projection Toward the $\mathbf{e}_2$ Direction

Projecting the column vector $\mathbf{x}$ onto the direction vector (unit vector) $\mathbf{e}_2$, the corresponding projection matrix $\mathbf{P}_2$ is

$$\mathbf{P}_2 = \mathbf{e}_2 @ \mathbf{e}_2^T = \begin{bmatrix} 0 \\ 1 \end{bmatrix} @ \begin{bmatrix} 0 \\ 1 \end{bmatrix}^T = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

As Figure 4 shows, the projection result corresponds to the following matrix multiplication

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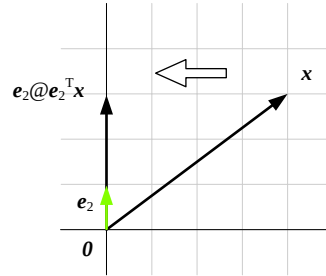
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$$\mathbf{P}_2 \mathbf{x} = \underbrace{\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}}_{\mathbf{e}_2 @ \mathbf{e}_2^T} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ x_2 \end{bmatrix}$$

Figure 4. Orthogonal Projection of the Two-Dimensional Column Vector $\mathbf{x}$ Onto the $\mathbf{e}_2$ Direction

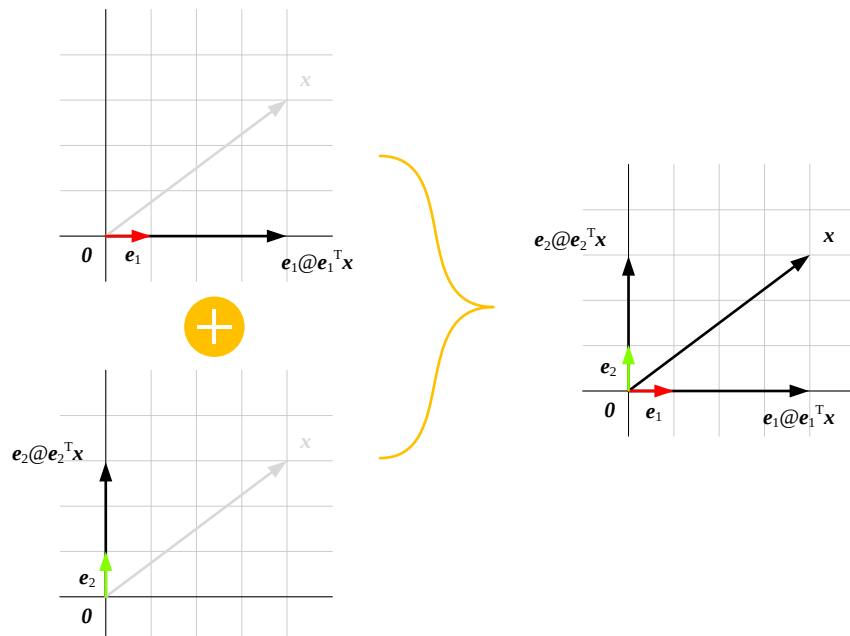
Vector Composition

and are each orthogonal projections that reduce information on their own; whereas and combined together exactly reconstruct all the original information, i.e.,

$$\mathbf{P}_1 \mathbf{x} + \mathbf{P}_2 \mathbf{x} = (\mathbf{P}_1 + \mathbf{P}_2) \mathbf{x} = (\mathbf{e}_1 @ \mathbf{e}_1^T + \mathbf{e}_2 @ \mathbf{e}_2^T) \mathbf{x} = \mathbf{I} @ \mathbf{x}$$

As Figure 5 shows, we have recovered !

In $[\mathbf{e}_1, \mathbf{e}_2]$, $\mathbf{x}$ performs an orthogonal decomposition of the vector; conversely, vector addition amounts to composing and recovering $\mathbf{x}$.

Figure 5. Recovering the Two-Dimensional Column Vector $\mathbf{x}$

Composing the Identity Matrix

In we also find an equation

$$\mathbf{e}_1 @ \mathbf{e}_1^T + \mathbf{e}_2 @ \mathbf{e}_2^T = \mathbf{I}$$

Where does this equation come from?

Let's look at an even "stranger" matrix multiplication

$$\mathbf{I} @ \mathbf{I}^T = \mathbf{I}_{2 \times 2}$$

Writing the unit vectors as row vectors, i.e., $[\mathbf{e}_1, \mathbf{e}_2]$, expanding the equation above gives

$$[\mathbf{e}_1 \quad \mathbf{e}_2] @ \begin{bmatrix} \mathbf{e}_1^T \\ \mathbf{e}_2^T \end{bmatrix} = \mathbf{e}_1 @ \mathbf{e}_1^T + \mathbf{e}_2 @ \mathbf{e}_2^T = \mathbf{I}$$

This actually uses the second perspective of matrix multiplication — the outer-product perspective.

As Figure 6 shows, we have in fact recovered the identity matrix.

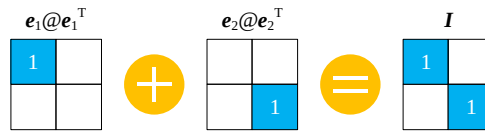


Figure 6. Recovering the Identity Matrix

Let's look at another equally "strange" matrix multiplication

$$\mathbf{I}^T @ \mathbf{I} = \mathbf{I}_{2 \times 2}$$

Writing the unit vectors as row vectors, i.e., $[\mathbf{e}_1, \mathbf{e}_2]$, expanding the equation above

$$\begin{bmatrix} \mathbf{e}_1^T \\ \mathbf{e}_2^T \end{bmatrix} @ [\mathbf{e}_1 \quad \mathbf{e}_2] = \begin{bmatrix} \mathbf{e}_1^T @ \mathbf{e}_1 & \mathbf{e}_1^T @ \mathbf{e}_2 \\ \mathbf{e}_2^T @ \mathbf{e}_1 & \mathbf{e}_2^T @ \mathbf{e}_2 \end{bmatrix} = \begin{bmatrix} \mathbf{e}_1 \cdot \mathbf{e}_1 & \mathbf{e}_1 \cdot \mathbf{e}_2 \\ \mathbf{e}_2 \cdot \mathbf{e}_1 & \mathbf{e}_2 \cdot \mathbf{e}_2 \end{bmatrix} = \begin{bmatrix} \|\mathbf{e}_1\|_2^2 & 0 \\ 0 & \|\mathbf{e}_2\|_2^2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \mathbf{I}$$

The equation above uses the first perspective of matrix multiplication — the inner-product perspective.

is the Gram matrix of the identity matrix $\mathbf{I}$. The main-diagonal entries are the inner products of $\mathbf{e}_1$ and $\mathbf{e}_2$ with themselves, i.e., the squared L^2 norm; the off-diagonal entries are the pairwise inner products of the vectors.

At this point, we have seen the "charm" of the standard orthonormal basis! We have also seen the wide applicability of the Gram matrix.

The basis vectors of the standard orthonormal basis are all unit vectors and are pairwise orthogonal; more importantly, the grid they form is the most natural unit-square grid.

Combining and gives

$$\mathbf{I}_{2 \times 2} @ \mathbf{I}_{2 \times 2}^T = \mathbf{I}_{2 \times 2}^T @ \mathbf{I}_{2 \times 2} = \mathbf{I}_{2 \times 2}$$

This tells us that the identity matrix is also an orthogonal matrix!

can also be written as

$$\begin{aligned} \mathbf{e}_1 @ \mathbf{e}_1^T &= \mathbf{I} - \mathbf{e}_2 @ \mathbf{e}_2^T \\ \mathbf{e}_2 @ \mathbf{e}_2^T &= \mathbf{I} - \mathbf{e}_1 @ \mathbf{e}_1^T \end{aligned}$$

? Does this remind you of how, in orthogonal projection, the projection matrix is defined using the normal vector?

The equation above tells us that in the plane, the horizontal line $x_2 = 1$ has corresponding normal vector $\mathbf{e}_2$, while the vertical line $x_1 = 1$ has corresponding normal vector $\mathbf{e}_1$.

This point is not very easy to see clearly in the plane; the next section will make it clearer using the 3×3 identity matrix.

Moreover, as Figure 7 shows, $\text{span}(\mathbf{e}_1)$ and $\text{span}(\mathbf{e}_2)$ are orthogonal complements of each other. $\text{span}(\mathbf{e}_1)$ and $\text{span}(\mathbf{e}_2)$ are both one-dimensional spaces. Simply put, the orthogonal complement contains all the vectors perpendicular to the original space.

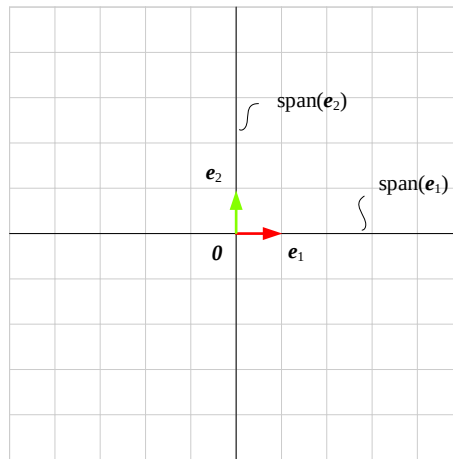


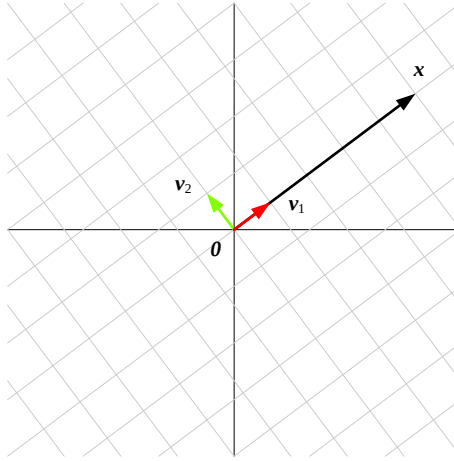
Figure 7. $\text{span}(\mathbf{e}_1)$ and $\text{span}(\mathbf{e}_2)$ Are Orthogonal Complements

2 × 2 Rotation Matrices

This book mentioned earlier that rotation matrices are also orthogonal matrices. Let's first look at the 2×2 rotation matrix $\mathbf{V}$, with a specific example being

$$\mathbf{V} = \begin{bmatrix} 4/5 & -3/5 \\ 3/5 & 4/5 \end{bmatrix} = \begin{bmatrix} 4/5 \\ 3/5 \end{bmatrix}_{\mathbf{v}_1} \begin{bmatrix} -3/5 \\ 4/5 \end{bmatrix}_{\mathbf{v}_2}$$

As Figure 8 shows, the vector $\mathbf{x}$ also has its own coordinates in the orthonormal basis $[\mathbf{v}_1, \mathbf{v}_2]$.

Figure 8. The Plane Spanned by Linear Combinations of $[v_1, v_2]$

Composing the Identity Matrix

Since V is an orthogonal matrix, it satisfies

$$VV^T = V^TV = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$

Let's first look at the Gram matrix V^TV , writing V as $[v_1, v_2]$, and expanding using the first perspective of matrix multiplication

$$\begin{bmatrix} v_1^T \\ v_2^T \end{bmatrix} @ [v_1 \ v_2] = \begin{bmatrix} v_1^T @ v_1 & v_1^T @ v_2 \\ v_2^T @ v_1 & v_2^T @ v_2 \end{bmatrix} = \begin{bmatrix} v_1 \cdot v_1 & v_1 \cdot v_2 \\ v_2 \cdot v_1 & v_2 \cdot v_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I_{2 \times 2}$$

Please analyze on your own the characteristics of the main-diagonal and off-diagonal entries of the square matrix in .

Now let's look at VV^T , expanding using the second perspective of matrix multiplication

$$[v_1 \ v_2] @ \begin{bmatrix} v_1^T \\ v_2^T \end{bmatrix} = v_1 @ v_1^T + v_2 @ v_2^T = I_{2 \times 2}$$

What we see is the superposition of two orthogonal projection matrices, and the result is the identity matrix.

Orthogonal Projection Toward the v_1 Direction

Projecting the two-dimensional column vector x onto the direction vector (unit vector) v_1 , the corresponding projection matrix P_1 is

$$P_1 = v_1 @ v_1^T = \begin{bmatrix} 4/5 \\ 3/5 \end{bmatrix} @ \begin{bmatrix} 4/5 \\ 3/5 \end{bmatrix}^T = \begin{bmatrix} 16/25 & 12/25 \\ 12/25 & 9/25 \end{bmatrix}$$

The projection result corresponds to the following matrix multiplication

$$\mathbf{P}_1 \mathbf{x} = \underbrace{\begin{bmatrix} 16/25 & 12/25 \\ 12/25 & 9/25 \end{bmatrix}}_{\mathbf{v}_1 @ \mathbf{v}_1^T} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 16x_1/25 + 12x_2/25 \\ 12x_1/25 + 9x_2/25 \end{bmatrix}$$

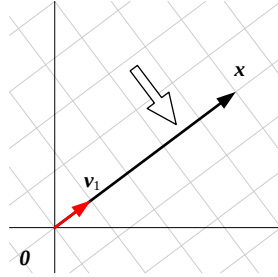


Figure 9. Orthogonal Projection of the Two-Dimensional Column Vector $\mathbf{x}$ Onto the $\mathbf{v}_1$ Direction

Figure 9 shows the orthogonal projection of the two-dimensional column vector $\mathbf{x}$ onto the $\mathbf{v}_1$ direction. Substituting specific values, the projection result is

$$\mathbf{P}_1 \mathbf{x} = \underbrace{\begin{bmatrix} 16/25 & 12/25 \\ 12/25 & 9/25 \end{bmatrix}}_{\mathbf{v}_1 @ \mathbf{v}_1^T} \begin{bmatrix} 4 \\ 3 \end{bmatrix} = \begin{bmatrix} 4 \\ 3 \end{bmatrix}$$

? Are you perhaps puzzled about why the orthogonal projection of the vector $\mathbf{x}$ onto $\mathbf{v}_1$ gives $\mathbf{x}$ itself? The reason is simple: $\mathbf{x}$ is parallel to $\mathbf{v}_1$, meaning that $\mathbf{x}$ is in $\text{span}(\mathbf{v}_1)$.

To compute the coordinate of $\mathbf{x}$ in $\text{span}(\mathbf{v}_1)$, we can simply use scalar projection, i.e.,

$$\mathbf{v}_1^T \mathbf{x} = \begin{bmatrix} 4/5 \\ 3/5 \end{bmatrix}^T \begin{bmatrix} 4 \\ 3 \end{bmatrix} = 5$$

This scalar value is equivalent to the coordinate of the vector $\mathbf{x}$ along $\mathbf{v}_1$; multiplying it by $\mathbf{v}_1$ is equivalent to "re-projecting" it back onto the plane.

Orthogonal Projection Toward the $\mathbf{v}_2$ Direction

Projecting the two-dimensional column vector $\mathbf{x}$ onto the direction vector (unit vector) $\mathbf{v}_2$, the corresponding projection matrix $\mathbf{P}_2$ is

$$\mathbf{P}_2 = \mathbf{v}_2 @ \mathbf{v}_2^T = \begin{bmatrix} -3/5 \\ 4/5 \end{bmatrix} @ \begin{bmatrix} -3/5 \\ 4/5 \end{bmatrix}^T = \begin{bmatrix} 9/25 & -12/25 \\ -12/25 & 16/25 \end{bmatrix}$$

The projection result corresponds to the following matrix multiplication

$$\mathbf{P}_2 \mathbf{x} = \underbrace{\begin{bmatrix} 9/25 & -12/25 \\ -12/25 & 16/25 \end{bmatrix}}_{\mathbf{v}_2 @ \mathbf{v}_2^T} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 9x_1/25 - 12x_2/25 \\ -12x_1/25 + 16x_2/25 \end{bmatrix}$$

Figure 10 shows the orthogonal projection of the two-dimensional column vector $\mathbf{x}$ onto the $\mathbf{v}_2$ direction. Substituting specific values, the projection result is

$$\mathbf{P}_2 \mathbf{x} = \underbrace{\begin{bmatrix} 9/25 & -12/25 \\ -12/25 & 16/25 \end{bmatrix}}_{\mathbf{v}_2 @ \mathbf{v}_2^T} \begin{bmatrix} 4 \\ 3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Since $\mathbf{x}$ and $\mathbf{v}_2$ are orthogonal, $\mathbf{x}$'s orthogonal projection onto $\mathbf{v}_2$ is the zero vector $\mathbf{0}$.

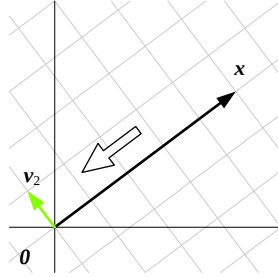


Figure 10. Orthogonal Projection of the Two-Dimensional Column Vector $\mathbf{x}$ Onto the $\mathbf{v}_2$ Direction

Vector Composition

As Figure 11 shows, superimposing and recovers $\mathbf{x}$ in full

$$(\mathbf{P}_1 + \mathbf{P}_2) \mathbf{x} = (\mathbf{v}_1 @ \mathbf{v}_1^T + \mathbf{v}_2 @ \mathbf{v}_2^T) \mathbf{x} = \begin{bmatrix} 16x_1/25 + 12x_2/25 \\ 12x_1/25 + 9x_2/25 \end{bmatrix} + \begin{bmatrix} 9x_1/25 - 12x_2/25 \\ -12x_1/25 + 16x_2/25 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

As Figure 12 shows, $\text{span}(\mathbf{v}_1)$ and $\text{span}(\mathbf{v}_2)$ are orthogonal complements of each other.

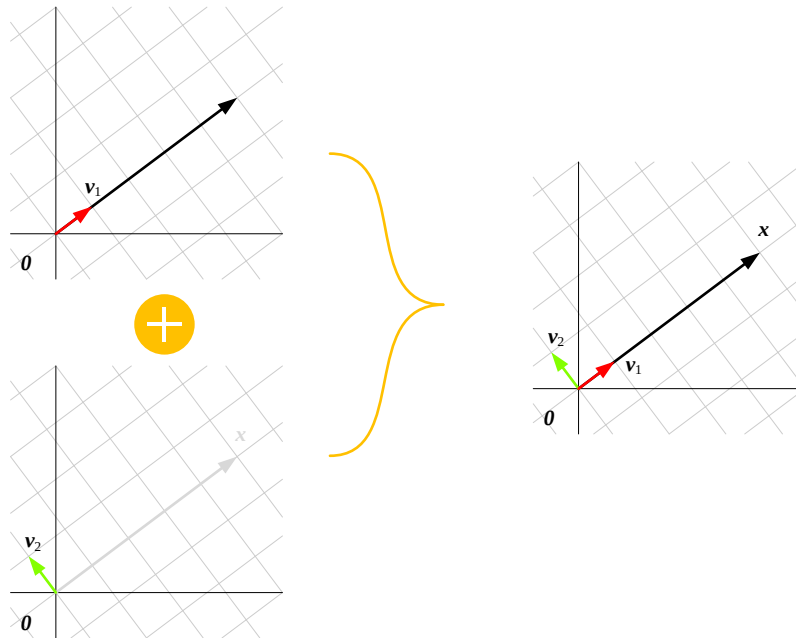
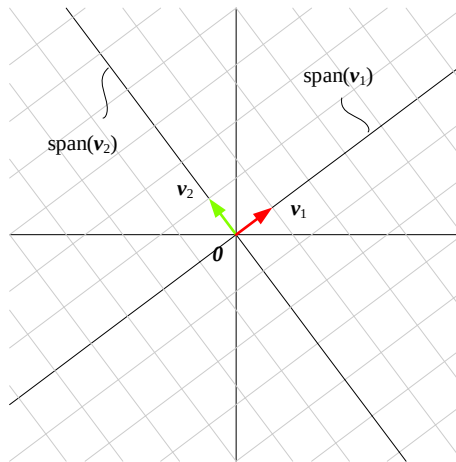


Figure 11. Recovering the Two-Dimensional Column Vector in $[\mathbf{v}_1, \mathbf{v}_2] \mathbf{x}$

Figure 12. $\text{span}(v_1)$ and $\text{span}(v_2)$ Are Orthogonal Complements

In the next section we raise the dimension, continuing to analyze 3×3 orthogonal matrices from a three-dimensional geometric perspective.



Please use tools such as DeepSeek/ChatGPT to complete the following exercises for this section.

Q1. Determine which of the following matrices are orthogonal matrices.

▶ $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

▶ $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$

▶ $\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$

▶ $\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$

▶ $\begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$

▶ $\begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$

▶ $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

▶ $\begin{bmatrix} \sqrt{2}/2 & \sqrt{2}/2 \\ \sqrt{2}/2 & \sqrt{2}/2 \end{bmatrix}$

▶ $\begin{bmatrix} \sqrt{2}/2 & -\sqrt{2}/2 \\ \sqrt{2}/2 & \sqrt{2}/2 \end{bmatrix}$

Q2. From a geometric point of view, what geometric transformations do the orthogonal matrices in Q1 correspond to?

Q3. Compute the determinants of the orthogonal matrices in Q1.

Q4. In each of the following matrix pairs, $\mathbf{A}$ and $\mathbf{B}$ are both orthogonal matrices. Determine whether $\mathbf{A@B}$ and $\mathbf{B@A}$ are both orthogonal matrices as well.

▶ $\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \mathbf{B} = \begin{bmatrix} \sqrt{2}/2 & -\sqrt{2}/2 \\ \sqrt{2}/2 & \sqrt{2}/2 \end{bmatrix}$

▶ $\mathbf{A} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \mathbf{B} = \begin{bmatrix} \sqrt{2}/2 & -\sqrt{2}/2 \\ \sqrt{2}/2 & \sqrt{2}/2 \end{bmatrix}$

▶ $\mathbf{A} = \begin{bmatrix} 3/5 & -4/5 \\ 4/5 & 3/5 \end{bmatrix}, \mathbf{B} = \begin{bmatrix} \sqrt{2}/2 & -\sqrt{2}/2 \\ \sqrt{2}/2 & \sqrt{2}/2 \end{bmatrix}$

Q5. From a geometric point of view, in Q4, what geometric transformations do $\mathbf{A@B}$ and $\mathbf{B@A}$ correspond to?

Q6. Compute the inverse and transpose of each of the following orthogonal matrices. What conclusion can you draw?

▶ $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

▶ $\begin{bmatrix} 3/5 & -4/5 \\ 4/5 & 3/5 \end{bmatrix}$

▶ $\begin{bmatrix} \sqrt{3}/2 & -1/2 \\ 1/2 & \sqrt{3}/2 \end{bmatrix}$

Q7. Compute the scalar projection and vector projection of the point $[2, 3]^T$ in the plane onto $\mathbf{e}_1$.

Q8. Compute the scalar projection and vector projection of the point $[2, 3]^T$ in the plane onto $\mathbf{e}_2$.

Q9. Project $[2, 3]^T$ onto $\mathbf{e}_1$ and $\mathbf{e}_2$ respectively, then combine the results.